

Structural Transformation and ESG Integration of Lotte Corporation (004990.KS): A Comprehensive Analysis of Sustainability Disclosures (2014–2023)

Company Identification and Business Profile

The entity associated with the Korea Exchange (KRX) ticker 004990.KS is Lotte Corporation, the preeminent holding company of the South Korean Lotte Group.¹ It is imperative to distinguish this entity from Lotte Holdings Co., Ltd., which is the group's Japanese counterpart headquartered in Shinjuku, Tokyo.¹ While the user prompt identifies the ticker as Lotte Holdings, verified financial data and official company profiles confirm that 004990.KS represents Lotte Corporation, a South Korean multinational conglomerate and the fifth-largest chaebol in the Republic of Korea.¹

The primary business activity of Lotte Corporation has evolved significantly over its history. Originally established in 1967 as Lotte Confectionery Co., Ltd., the company underwent a pivotal transition in October 2017 to become a dedicated holding company.² In its current capacity, Lotte Corporation serves as the strategic "control tower" for more than 90 business units.¹ These subsidiaries are engaged in a diverse array of sectors, including food manufacturing (Lotte Wellfood, Lotte Chilsung), retail (Lotte Shopping, Lotte Hi-Mart), industrial chemicals (Lotte Chemical, Lotte Fine Chemical), construction (Lotte E&C), and various service-oriented businesses including tourism and financial services.¹

The headquarters of Lotte Corporation is located at the Lotte World Tower in the Songpa District of Seoul, South Korea.¹ This landmark facility not only houses the executive operations of the holding company but also symbolizes the group's global ambitions. Lotte Corporation is classified under the Diversified Retail and Consumer Cyclical sectors in various financial indices, though its functional reality is that of an industrial conglomerate.³ This identification is corroborated by primary sources such as the Korea Exchange (KSC) listing, Lotte's official Investor Relations portal, and reputable secondary financial data providers including Bloomberg, Reuters, and Morningstar.³

Historical Context and the Evolution of Governance

To understand the ESG trajectory of Lotte Corporation, one must first examine the structural

reforms that occurred during the mid-to-late 2010s. For much of its early history, the Lotte Group operated through a complex web of circular shareholdings, which was a common but increasingly criticized feature of South Korean chaebols.⁵ This lack of transparency prompted significant regulatory pressure and internal reform initiatives aimed at professionalizing management and enhancing shareholder value.⁵

The 2017 transition to a holding company structure was the most significant milestone in this regard. By consolidating its diverse business units under Lotte Corporation, the group simplified its ownership structure and established a clear chain of command for ESG strategy.⁴ This transition was not merely administrative; it represented a fundamental shift in the company's philosophy. Celebrating its 50th anniversary in 2017, Lotte adopted the "Lifetime Value Creator" vision, signaling a move from quantitative growth toward qualitative development and long-term sustainability.⁴

This restructuring era was preceded by periods of significant external scrutiny. In 2016, the group faced a major corruption investigation, which eventually led to the 2018 bribery conviction of Chairman Shin Dong-bin.³ These controversies served as a powerful catalyst for the governance reforms that followed. The subsequent decade has been characterized by an aggressive effort to restore public and investor trust through the institutionalization of ethics, compliance, and transparent ESG reporting.⁸

ESG Report Timeline Analysis (2014–2023)

The decade from 2014 to 2023 represents the maturation of Lotte's sustainability disclosure from subsidiary-level environmental reporting to a unified, group-wide ESG framework led by Lotte Corporation.

The Era of Decentralized Reporting (2014–2016)

During this early phase, Lotte Corporation—then still operating primarily as Lotte Confectionery—did not yet publish a centralized sustainability report. Instead, ESG disclosure was led by its major listed subsidiaries, most notably Lotte Shopping and Lotte Chemical.⁴

In 2014, Lotte Shopping published a sustainability report prepared in accordance with the Global Reporting Initiative (GRI) G4.0 guidelines.¹¹ This report focused on its four major business divisions: Department Store, Mart, Super, and Cinema.¹¹ Notably, Lotte Shopping was a global leader in the retail sector during this time, having been selected for the Dow Jones Sustainability Indices (DJSI) World from 2009 to 2014 and serving as the Retail Industry Group Leader for four consecutive years.¹¹ The 2014 report emphasized "fair play-based" management and included third-party assurance from SolAbility.¹¹

Lotte Chemical also maintained a strong reporting cadence, having initiated annual ESG reports as early as 2007.¹⁰ These early disclosures were characterized by a focus on environmental

safety and emission metrics relevant to the petrochemical industry.¹⁰ Similarly, Lotte Engineering & Construction (E&C) maintained an active reporting history, with a 2014 report focusing on safety management and sustainable infrastructure.¹³

The Restructuring and Foundation Phase (2017–2020)

The year 2017 marked the reorganization of the group into its current holding company structure.⁴ Following this transition, Lotte Corporation began laying the groundwork for holding-level ESG oversight. In May 2018, Lotte Corporation became a participant in the United Nations Global Compact (UNGC), committing to ten principles across human rights, labor, environment, and anti-corruption.¹⁴ However, according to the UNGC portal, the entity was later delisted for a failure to communicate progress, a status that likely reflected the administrative challenges of the initial transition period.¹⁴

In 2020, the confectionery division—now a subsidiary under Lotte Corporation—published its first modern sustainability report.¹⁵ This document, covering the 2020–2021 period, was prepared according to GRI Standards and verified by the Korea Management Registrar.¹⁵ It introduced the concept of "Sweet ESG" and highlighted the company's efforts to achieve an A grade in the ESG evaluation by the Korea Institute of Corporate Governance and Sustainability (KCGS).¹⁵

Year	Milestone	Description	Source	Relevance
2014	Lotte Shopping DJSI Leadership	Named Retail Industry Group Leader for 4th consecutive year.	¹¹	Established Lotte's early credibility in sustainability.
2017	Holding Company Transition	Lotte Corporation established as the group's "control tower."	⁴	Simplified governance; enabled centralized ESG strategy.
2018	UN Global Compact Entry	Lotte Corporation joined as a participant.	¹⁴	Commitment to international human rights and labor norms.

2020	Lotte Confectionery 1st SR	First sustainability report published for the confectionery unit.	15	Initiated formalized reporting for the core food business.
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The ESG Management Declaration and Standardization (2021–2023)

July 2021 was a transformative month for the conglomerate. Lotte Group held an "ESG Management Declaration Ceremony," where Chairman Shin Dong-bin pledged to transform the group into a global leader in ESG management by 2025.⁸ This declaration triggered several immediate regulatory and disclosure milestones:

First, Lotte Corporation published its inaugural group-wide sustainability report in 2021.⁴ This report served as the first consolidated disclosure of the holding company's mid-to-long-term ESG strategies and the performance of its subsidiaries.⁴ Second, ESG Committees were established under the Board of Directors of all ten listed group affiliates.¹⁶ Third, the group pioneered the integration of ESG Key Performance Indicators (KPIs) into the performance evaluations of all group affiliate CEOs, ensuring that sustainability metrics had direct financial and career implications for top leadership.⁸

In 2022, Lotte Corporation advanced its climate strategy by announcing the "2040 Carbon Neutrality Goal" and establishing "Carbon Neutral Roadmap 1.0".⁹ The 2022 sustainability report included detailed TCFD (Task Force on Climate-related Financial Disclosures) and GRI indices, and was verified by third-party assurance statements to ensure data reliability.¹⁶

By 2023, the sophistication of reporting reached new levels. Lotte Shopping became the first Korean retail company to join the Science Based Targets initiative (SBTi) to verify its greenhouse gas reduction plans.¹⁷ Lotte Wellfood (the newly merged and renamed entity) joined the RE100 initiative, committing to 100% renewable electricity by 2040.¹⁸ Additionally, Lotte Corporation obtained the ISO 37301 certification for its Compliance Management System, marking a major milestone in governance transparency.⁹

Environmental Stewardship and Climate Strategy

Lotte Corporation's environmental strategy is framed by the recognition that its diversified operations—particularly in chemicals and retail—have significant ecological impacts. The "Carbon Neutral Roadmap 2.0," announced in 2023, is the central mechanism for the company's environmental transition.⁹

Net Zero 2040 and GHG Emission Targets

Lotte Group's 2040 Carbon Neutrality Goal is a bold commitment that precedes South Korea's national target by ten years.¹⁹ The strategy is bifurcated into short-term process efficiencies and long-term technological innovations. By 2030, the group aims to reduce greenhouse gas emissions at its business sites by 22% compared to 2018 levels.⁹ This target expands to a 61% reduction by 2040, ultimately reaching net zero.⁹

The chemical business unit, a major contributor to the group's carbon footprint, has set a 14.5% reduction target by 2030 compared to 2018.⁵ To achieve this, Lotte Chemical is transitioning its portfolio toward high-value-added specialty products and reducing its reliance on basic materials.⁵ In contrast, non-chemical businesses, including food and retail, have a more aggressive 35% reduction target for 2030.⁵

Environmental Metric	2022 Performance	2023 Performance	2024 Performance	Trend Analysis
Total GHG Emissions (Scope 1&2)	722,630 tCO ₂ eq	710,049 tCO ₂ eq	715,622 tCO ₂ eq	Gradual decrease followed by expansion-led stabilization. ²²
Direct Emissions (Scope 1)	101,259 tCO ₂ eq	96,042 tCO ₂ eq	96,484 tCO ₂ eq	Successful reduction in direct site emissions. ²²
Indirect Emissions (Scope 2)	621,371 tCO ₂ eq	614,007 tCO ₂ eq	619,138 tCO ₂ eq	Reflects heavy reliance on external grid electricity. ²²
Renewable Energy Generation (TJ)	N/A	20.9 TJ	33.7 TJ	61% year-over-year growth in solar PV capacity. ²²

The mechanism for achieving these targets involves six core reduction measures: the expansion of renewable energy through solar PV and Power Purchase Agreements (PPAs), the development of hydrogen energy as a clean fuel source, the implementation of Carbon

Capture, Utilization, and Storage (CCUS) technologies, the transition to zero-emission commercial vehicles, the enhancement of building energy efficiency, and the adoption of eco-friendly raw materials.⁹

Resource Circularity and Plastic Roadmaps

Lotte Chilsung Beverage and Lotte Wellfood have prioritized the "2030 Plastic Reduction Roadmap," which targets a 20% absolute reduction in plastic usage by 2030.⁵ This initiative has led to the widespread adoption of "no-label" water bottles and the lightweighting of beverage containers.⁵ Lotte Chemical is supporting these efforts through its "Green Promise 2030" strategy, which involves developing chemically recycled plastic materials that can be reintegrated into the food packaging supply chain.²⁴

Water management is another critical environmental focus. Lotte Shopping first declared environmental management in 1994 and has since participated in projects like the restoration of Seokchon Lake's water quality.⁹ Lotte Chemical has set a target to achieve 100% water recycling by 2050, demonstrating a long-term commitment to resource circularity.²⁶

Social Responsibility and Human Capital

The social pillar of Lotte Corporation's ESG strategy is anchored in its 2013 Diversity Charter and the 2022 Human Rights Statement.¹⁶ These documents provide the framework for managing a global workforce of 60,000 people across diverse cultural and regulatory environments.¹

Diversity, Equity, and Inclusion (DEI)

Lotte was the first major Korean conglomerate to declare a Diversity Charter, emphasizing respect for gender, culture, physical ability, and generation.¹⁶ The impact of this early adoption is evident in the company's internal metrics. Over the decade ending in 2023, the number of female employees in the group more than doubled, and the number of female executives increased fivefold.²⁷

In 2023, Lotte Group further institutionalized these values by establishing a DEI management system and disclosing its "Global HR Conference" results, which focused on the inclusion of the 49,000 local employees working in its overseas subsidiaries.²⁷ Lotte Shopping has set specific targets for 2025, aiming for female leadership ratios of up to 25% in its e-commerce division.²²

Supply Chain Management and Human Rights

The "Lotte Group Statement on Human Rights," enacted in 2022, is based on international principles such as the UN Universal Declaration of Human Rights and the ILO Declaration on Fundamental Principles and Rights at Work.²² This statement mandates that human rights risks

be identified and managed throughout the supply chain.²⁷

Lotte Wellfood and Lotte Shopping have initiated mandatory ESG audits for their key suppliers.¹⁷ The "Code of Conduct for Partners" outlines strict requirements for labor standards, including the prohibition of child labor and the guarantee of legal minimum wages.²² Lotte Shopping's partner self-assessment framework includes a detailed scoring system for labor hour management, ensuring that workers record and sign off on their own hours before wages are calculated.²²

Shared Growth and Community Engagement

Lotte Corporation facilitates "shared growth" through various SME support programs. The "Lotte-Republic of Korea Brand Expo," which debuted in Thailand in 2018, has supported nearly 1,500 small businesses in their international expansion efforts.⁹ In the first half of 2024, Lotte Shopping hosted its "Lotte Happiness Sharing Concert" for partner company employees, demonstrating a holistic approach to stakeholder engagement.⁹

Governance and Ethics Reform

The governance landscape for Lotte Corporation has been redefined by the 2017 holding company transition and the subsequent professionalization of its Board of Directors.⁴

Board Structure and Independence

As the holding company, Lotte Corporation has mandated that all listed subsidiaries establish ESG Committees under their respective Boards of Directors.¹⁶ These committees are tasked with reviewing sustainability strategies and ensuring that ESG performance is reflected in executive compensation.⁸ By the end of 2023, these committees were predominantly composed of independent non-executive directors with expertise in risk management, finance, and environmental law.¹⁸

The group has also adopted shareholder-friendly policies, including electronic voting and the transparent disclosure of corporate information through IR Days.¹⁷ Lotte Wellfood, for instance, committed to gradually increasing its dividend payout ratio to over 30% following its merger with Lotte Foods in 2022.²⁹

Compliance and Anti-Corruption

One of the most significant governance milestones in the 2022–2023 period was the adoption of the ISO 37301 Compliance Management System.⁹ Lotte Corporation received this certification from the Korea Management Certification Institute in September 2023.⁹ This standard requires the group to systematize risk identification, assessment, control, monitoring,

and remediation procedures.⁹

To support this, the group holds an annual "Compliance Forum" and establishes guidelines to prevent bribery and corruption.⁹ Subsidiary units such as Lotte E&C have successfully obtained ISO 37001 (Anti-Corruption Management System) and ISMS-P (Personal Information & Privacy Protection) certifications.¹⁹

External ESG Assessments and Ratings

External rating agencies provide a critical validation of Lotte Corporation's ESG efforts. The general trend over the last three years has been one of improvement, particularly in the Governance and Social categories.

KCGS and MSCI Ratings

The Korea Institute of Corporate Governance and Sustainability (KCGS) has consistently assigned Lotte Corporation high marks in recent years. For the 2023 and 2024 fiscal years, Lotte Corporation maintained an integrated rating of "A," with its Social (S) category achieving the highest possible "A+" grade.³⁰

Rating Agency	2021 Rating	2022 Rating	2023 Rating	2024 Rating
KCGS Integrated	A	A	A	A
MSCI Integrated	BBB	BBB	BBB	A
Sustainalytics Risk	N/A	N/A	N/A	14.22 (Low)

Lotte Chemical's performance in the MSCI ESG evaluation is particularly noteworthy, receiving an "A" rating for two consecutive years (2024–2025).²⁵ The company was recognized for its tangible achievements in carbon neutrality and the development of a "green ecosystem".²⁵

Sustainalytics and ESG Risk Management

As of March 2026, Morningstar Sustainalytics categorized Lotte Corporation (004990) as **Low Risk**, with a quantitative score of **14.22**.³² This score indicates that the company’s unmanaged ESG risk is minimal relative to its global peers.³² While Sustainalytics identified "Business Ethics" and "Customer Incidents" as potential areas of controversy, the overall management of these risks was deemed effective, leading to a high "Global Universe" ranking.³²

CDP and S&P Global

Lotte Chemical has demonstrated consistent participation in the Carbon Disclosure Project (CDP), receiving a "B" grade for Climate and an "A-" for Water Resources in 2023.³³ S&P Global’s Corporate Sustainability Assessment (CSA) currently assigns Lotte Corporation an ESG score of 53, placing it within the mid-range of industrial conglomerates, though this score is primarily based on publicly available modeling rather than active participation in the full CSA questionnaire.³⁴

Financial Implications of ESG Strategy

The integration of ESG into Lotte Corporation’s business model is increasingly linked to its market valuation and investment appeal. As of early 2026, Lotte’s market capitalization reached approximately \$2.19 trillion KRW, representing a significant recovery from the volatility seen during the 2018–2020 period.³⁶

Analysts have noted that Lotte’s "Turnaround" status is supported by its improving ESG quality scores.⁶ The company’s Price-to-Book (P/B) ratio of 0.39 and Price-to-Sales (P/S) ratio of 0.14 suggest that it is currently undervalued by the market relative to its assets and revenue, potentially due to the legacy of its past governance complexities.⁶ However, the systematic reduction of ESG risk, as evidenced by the Sustainalytics Low Risk rating, is expected to narrow this valuation gap over time.⁶

Furthermore, Lotte Corporation’s inclusion in several ESG-screened ETFs, such as the iShares MSCI EM IMI ESG Screened UCITS ETF, confirms that its sustainability disclosures now meet the rigorous criteria of international institutional investors.³⁷

Reference Table with Credibility Assessment

#	Reference (APA 7th edition format)	Source Type	Credibility Score (0–10)	Justification for Score	Relevance to Query

1	Lotte Corporation. (2024). <i>2024 Sustainability Report</i> .	Primary	10	Official company disclosure filed with regulators and audited by third parties.	Primary source for current ESG KPIs, roadmaps, and data.
2	KCGS. (2024). <i>ESG Ratings for Listed Companies</i> .	Secondary	9	Leading domestic agency with transparent methodology.	Key source for domestic benchmarking and historical ratings.
3	MSCI ESG Research. (2025). <i>Corporate Ratings: Lotte Corp</i> .	Secondary	9	Globally recognized investment research standard.	Verification of international ESG standing and performance trends.
4	Morningstar Sustainalytics . (2026). <i>ESG Risk Rating: 004990.KS</i> .	Secondary	8	Independent risk assessor; provides quantitative risk metrics.	Detailed data on unmanaged ESG risks and controversies.
5	UN Global Compact. (2024). <i>Participant Profile: Lotte</i>	International	9	Official status record from an international intergovernm	Verification of commitment dates and communication status.

	Corporation.			ental body.	
6	Reuters. (2018, February 21). <i>Lotte Group chairman relinquishes rights.</i>	Secondary	7	Reputable financial news with established editorial standards.	Contextual information on governance controversies.
7	Lotte Shopping. (2014). <i>Sustainability Report 2014.</i>	Primary	10	Historical record for the 2014 timeline.	Base data for early GRI G4 adoption.
8	S&P Global. (2025). <i>Corporate Sustainability Assessment Results.</i>	Secondary	8	Major global index provider.	Benchmarking against global industrial conglomerates.

Transparency and Limitations

This analysis is based on the comprehensive research material available as of March 2026. The analyst acknowledges several limitations inherent in the data collection process:

The reporting scope of Lotte Corporation as a holding company sometimes involves consolidated figures that may mask the specific performance of smaller, unlisted subsidiaries. While the group-level "Carbon Neutrality Roadmap" provides a high-level view, the specific emission trajectories for non-listed units are often less transparent than those for flagship entities like Lotte Chemical or Lotte Shopping.⁵

The status of the company's relationship with the UN Global Compact contains a historical discrepancy. While snippet ¹⁴ indicates the company was delisted for "failure to communicate

progress," more recent reports ⁴ emphasize adherence to the Ten Principles. This likely suggests a past administrative lapse during the 2017 restructuring that has since been addressed, though a formal "re-commitment" status may still be in transition.

Finally, some real-time financial metrics and deep-dive analyst forecasts from paywalled platforms like Bloomberg or Refinitiv were accessed only via summarized snippets. Users are encouraged to independently verify these figures through the following authoritative portals:

1. **KIND (Korea Investor's Network for Disclosure System):** The official repository for all mandatory KRX filings [<https://kind.krx.co.kr>].
2. **Lotte Corporation IR Archive:** The definitive source for annual sustainability and annual reports [<https://www.lotte.co.kr/global/en/ir/archive.do>].
3. **CDP Disclosure Search:** For detailed climate and water resource responses [<https://www.cdp.net/en/responses>].

Conclusion and Future Strategic Outlook

The analysis of Lotte Corporation (004990.KS) throughout the 2014–2023 period reveals a company that has successfully navigated a complex transition from a non-transparent conglomerate to a modern, ESG-integrated holding company. The 2017 restructuring served as the essential governance foundation, while the 2021 ESG Declaration provided the operational momentum for standardized reporting and executive accountability.⁸

Looking ahead, Lotte Corporation faces the challenge of translating its "Net Zero 2040" ambitions into realized emission reductions amid a volatile global energy market.⁹ Its proactive adoption of ISSB and CSRD-aligned reporting suggests a readiness for the next generation of global disclosure mandates.⁵ Provided the group maintains its rigorous compliance framework and continues to foster diversity and human rights across its international business units, it is well-positioned to maintain its "Low Risk" standing and enhance its long-term corporate value.

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